6406533515784. ✖ Rs. 1,69,00,000/-

Sw Testing

Section Id :	64065375418
Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	17
Number of Questions to be attempted :	17
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653160190
Question Shuffling Allowed :	No

Question Number : 32 Question Id : 6406531040745 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SOFTWARE TESTING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533515817. ✔ YES

6406533515818. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653160191
Question Shuffling Allowed :	Yes

Question Number : 33 Question Id : 6406531040746 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Which type of testing involves testing the limit of the system, that is, the magnitude of demand that can be placed on the system while continuing to meet its latency, response time and throughput requirements?

Options :

6406533515819. ✖ Reliability testing

6406533515820. ✔ Scalability testing

6406533515821. ✖ Compatibility testing

6406533515822. ✖ Regulatory testing

Question Number : 34 Question Id : 6406531040752 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Which client-side testing needs to verify the inter-value constraints like credit card number and the expiry date?

Options :

6406533515839. ✖ Value level bypass testing

6406533515840. ✔ Parameter level bypass testing

6406533515841. ✖ Control flow level bypass testing

6406533515842. ✖ User-session Data based Testing

Question Number : 35 Question Id : 6406531040753 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider transitions between two web pages using back button and forward button, which have to be represented in an Application Transition Graph (ATG). Which type of transitions are these?

Options :

6406533515843. ✔ Operational transition

6406533515844. ✖ Redirect transition

6406533515845. ✖ Form link transition

6406533515846. ✖ Component expression transition

Question Number : 36 Question Id : 6406531040766 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider four partitions as $[X_1, X_2]$, $[Y_1, Y_2, Y_3, Y_4]$, $[W_1, W_2, W_3]$, and $[Z_1, Z_2]$. What is the minimum number of test cases required to satisfy Each Choice Coverage (ECC) criteria?

Options :

6406533515887. ✔ 4

6406533515888. ✖ 7

6406533515889. ✖ 8

6406533515890. ✖ 48

Sub-Section Number : 3

Sub-Section Id : 640653160192

Question Shuffling Allowed : Yes

Question Number : 37 Question Id : 6406531040747 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are *true* about software quality assurance activities?

Options :

6406533515823. ✔ The activities establish and evaluate the processes that produce products.

6406533515824. ✖ The activities focus on determining the defects in the actual products produced.

6406533515825. ✖ Involves product-focused action.

6406533515826. ✔ Involves process-focused action.

Question Number : 38 Question Id : 6406531040765 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following predicate(s) is/are in Disjunctive Normal Form (DNF) ?

Options :

6406533515883. ✔ $(a \wedge b) \vee b \vee c$

6406533515884. ✖ $(a \wedge b) \wedge c$

6406533515885. ✔ $(a \wedge b) \vee (c \wedge d) \vee e$

6406533515886. ✖ $(a \vee b) \wedge (c \vee d)$

Sub-Section Number : 4

Sub-Section Id : 640653160193

Question Shuffling Allowed : Yes

Question Number : 39 Question Id : 6406531040748 Question Type : MSQ

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider test-driven development (TDD) for a web portal for employees of an organization. The employees can claim for TA (travel allowances) through this portal. One of the user stories of the web portal is "TA amount validation: Invalid TA amount can be verified." Below are the test cases for the "TA amount validation" user story.

```
public class TestTAValidation {
    private TAValidation taObj;
    @Test
    public void test() {
        taObj = new TAValidation();
        assertEquals(taObj.isValid(100), true);
        assertEquals(taObj.isValid(500), true);
        assertEquals(taObj.isValid(1000), true);
        assertEquals(taObj.isValid(99), false);
        assertEquals(taObj.isValid(1001), false);
    }
}
```

Which of the following are the appropriate implementations for the user story?

Options :

```
public class TAValidation {
    public boolean isValid(int ta) {
        if(ta < 100) {
            System.out.println("TA claimed is less than allowed amount");
            return false;
        }
        else if(ta > 1000) {
            System.out.println("TA claimed is more than allowed amount");
            return false;
        }
        else
            return true;
    }
}
```

6406533515827. ✔

6406533515828. ✖

```java
public class TAValidation {
    public boolean isValid(int ta) {
        if(ta <= 100) {
            System.out.println("TA claimed is less than allowed amount");
            return false;
        }
        else if(ta >= 1000) {
            System.out.println("TA claimed is more than allowed amount");
            return false;
        }
        else
            return true;
    }
}
```

```java
public class TAValidation {
    public boolean isValid(int ta) {
        if(ta <= 99) {
            System.out.println("TA claimed is less than allowed amount");
            return false;
        }
        else if(ta >= 1001) {
            System.out.println("TA claimed is more than allowed amount");
            return false;
        }
        else
            return true;
    }
}
```

6406533515829. ✔

```java
public class TAValidation {
    public boolean isValid(int ta) {
        if(ta > 100) {
            System.out.println("TA claimed is less than allowed amount");
            return false;
        }
        else if(ta < 1000) {
            System.out.println("TA claimed is more than allowed amount");
            return false;
        }
        else
            return true;
    }
}
```

6406533515830. ✖

Sub-Section Number : 5

Sub-Section Id : 640653160194

Question Shuffling Allowed : No

Question Id : 6406531040749 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (40 to 41)

Question Label : Comprehension

Consider the code segment of a Java servlet below. The atomic sections are marked as $P_1, P_2, P_3, \cdots$.

P_1	/*
	emp_salaries stores salaries disbursed each month during a financial year to each employee
	*/
	ArrayList<Double> emp_salaries = null;
	response.setContentType("text/html");
	PrintWriter out = response.getWriter();
	out.print("<HTML><HEAD><TITLE>");
	out.print("Employee Tax Calculator");
	out.println("</TITLE></HEAD><BODY> ");
	String emp_id = request.getParameter("EID");
	int year = request.getParameter("YEAR");
	/*
	getSalaries() considers the employee ID (emp_id) and financial year (year) as input, runs a query in database, and returns an ArrayList object containing the salaries of the employee during the given financial year
	*/
	emp_salaries = getSalaries(emp_id, year);
	double total_salary = 0.0;

	if(emp_salaries != null) {
	for (Double sal : emp_salaries) {
P_2	total_salary += sal;
	} if(total_salary >= 200000.00) {
P_3	double tax = total_salary * 0.1; out.println("Tax = " + tax +"</BR>");
	} else

P_4	{
	out.println("No tax applicable</BR>");
	}

	} else{
P_5	out.println("Invalid employee ID</BR>");
	}
P_6	out.println ("</BODY></HTML>"); out.close();

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 40 Question Id : 6406531040750 Question Type : MCQ
Correct Marks : 5
Question Label : Multiple Choice Question
Identify the component expression corresponding to the given code.
Options :

6406533515831. ✖ $P_1 \cdot (((P_2|P_3^*) \cdot P_4)|P_5) \cdot P_6$

6406533515832. ✖ $P_1 \cdot ((P_2^*|(P_3^*|P_4))|P_5) \cdot P_6$

6406533515833. ✖ $P_1 \cdot ((P_2 \cdot (P_3^*|P_4))|P_5) \cdot P_6$

6406533515834. ✔ $P_1 \cdot ((P_2^* \cdot (P_3|P_4))|P_5) \cdot P_6$

Question Number : 41 Question Id : 6406531040751 Question Type : MCQ
Correct Marks : 4
Question Label : Multiple Choice Question
Identify the content variable
in atomic section P_3.
Options :

6406533515835. ✖ `out.println`

6406533515836. ✔ `tax`

6406533515837. ✖ `double`

6406533515838. ✖ `"Tax = " and "</BR>"`

Sub-Section Number :	6
Sub-Section Id :	640653160195
Question Shuffling Allowed :	No

Question Id : 6406531040754 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (42 to 44)
Question Label : Comprehension

Consider the following control flow graph (CFG).

- $G = \{V, E\}$,
- Set of nodes, $V = \{1, 2, 3, 4, 5, 6, 7\}$,
- Set of edges, $E = \{(1, 2), (2, 3), (2, 4), (3, 2), (4, 5), (4, 6), (5, 7), (6, 7)\}$,
- Initial node, $V_0 = \{1\}$,
- Final node, $V_f = \{7\}$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 42 Question Id : 6406531040755 Question Type : MSQ

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

We have computed some of the prime paths, which are as follows:

1. $[1, 2, 4, 6, 7]$,
2. $[1, 2, 4, 5, 7]$,
3. $[3, 2, 4, 6, 7]$,
4. $[1, 2, 3]$,
5. $[3, 2, 3]$.

Identify the missing prime path(s).

Options :

6406533515847. ✔ [2,3,2]

6406533515848. ✖ [2, 3, 4, 6]

6406533515849. ✔ [3,2,4,5,7]

6406533515850. ✖ [1, 2, 3, 2, 4, 5, 6, 7]

Question Number : 43 Question Id : 6406531040756 Question Type : MSQ

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the set of test paths {[1, 2, 4, 5, 7], [1, 2, 4, 6, 7], [1, 2, 3, 2, 3, 2, 4, 5, 7]}. Which of the following statements is/are true about satisfying the coverage criteria on CFG *G*?

Options :

6406533515851. ✔ It satisfies node coverage.

6406533515852. ✔ It satisfies edge coverage.

6406533515853. ✔ It satisfies edge-pair coverage.

6406533515854. ✖ It satisfies prime path coverage.

Question Number : 44 Question Id : 6406531040757 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What is the cyclomatic complexity of the CFG *G*?

Options :

6406533515855. ✖ 2

6406533515856. ✔ 3

6406533515857. ✖ 4

6406533515858. ✖ 5

Sub-Section Number :	7
Sub-Section Id :	640653160196
Question Shuffling Allowed :	No

Question Id : 6406531040758 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (45 to 47)

Question Label : Comprehension

Consider the following Java class below with a mutant indicated in the comments.

Answer the subquestions related to the given mutant in the method BinarySearch.

```
class BinarySearch {
    // Returns index of x if it is present in arr[],
    // else return -1
    int binarySearch(ArrayList<Integer> arr, int x) {
        int left = 0, right = arr.size() - 1;
        while (left <= right) {   // Mutant-1: while (left < right)
            int mid = left + (right - left) / 2;
            if (arr.get(mid) == x)
                return mid;
            else if (arr.get(mid) < x)
                left = mid + 1;
            else
                right = mid - 1;
        }
        return -1;
    }
    if (result == -1)
        System.out.println("Element not present");
    else
        System.out.println("The Element" + x + "is found at" + "index " + result);
}
```

Sub questions

Question Number : 45 Question Id : 6406531040759 Question Type : MCQ Correct Marks : 4

Question Label : Multiple Choice Question

Identify the mutation operator that has been applied at Mutant-1.

Options :

6406533515859. ✔ Relational Operator Replacement

6406533515860. ✖ Arithmetic Operator Replacement

6406533515861. ✖ Absolute Value Insertion

6406533515862. ✖ Conditional Operator Replacement

Question Number : 46 Question Id : 6406531040760 Question Type : MSQ
Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Identify the test cases which will not infect the statement of the mutant.

Options :

6406533515863. ✔ `arr = [], x = 0`

6406533515864. ✖ `arr = [0], x = 1`

6406533515865. ✔ `arr = [0, 1], x = 1`

6406533515866. ✔ `arr = [0, 1, 2] x = 2`

Question Number : 47 Question Id : 6406531040761 Question Type : MCQ
Correct Marks : 4

Question Label : Multiple Choice Question

Which of the following test cases would strongly kill the mutant?

Options :

6406533515867. ✖ `arr = [], x = 1`

6406533515868. ✔ `arr = [0], x = 0`

6406533515869. ✖ `arr = [0, 1], x = 1`

6406533515870. ✖ `arr = [0, 1, 2], x = 0`

Sub-Section Number :	8
Sub-Section Id :	640653160197
Question Shuffling Allowed :	No

Question Id : 6406531040767 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (48 to 49)

Question Label : Comprehension

Consider the truth table for predicate $p = a \vee (\neg b \wedge \neg c)$ given below.
Each blank cell in truth table represents a F.

Row#	a	b	c	p	p_a	p_b	p_c
1	T	T	T	T	T		
2	T	T		T	T		
3	T		T	T	T		
4	T			T			
5		T	T		T		
6		T			T	T	
7			T		T		
8				T		T	T

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 48 Question Id : 6406531040768 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Which of the following represents p_a?

Options :

6406533515891. ✖ $(\neg c \wedge \neg b)$

6406533515892. ✖ $(c \wedge b)$

6406533515893. ✖ $(\neg c \vee \neg b)$

6406533515894. ✔ $(c \vee b)$

Question Number : 49 Question Id : 6406531040769 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

With reference to the truth table row numbers (Row#) given in the main data, identify the set of all valid row pairs to satisfy RACC for clause *a*?

Options :

6406533515895. ✖ {(7, 8)}

6406533515896. ✖ {(6, 8)}

6406533515897. ✔ {(1, 5), (2, 6), (3, 7)}

6406533515898. ✖ {(1, 5), (1, 6), (1, 7), (2, 5), (2, 6), (2, 7), (3, 5), (3, 6), (3, 7)}

Sub-Section Number : 9

Sub-Section Id : 640653160198

Question Shuffling Allowed : Yes

Question Number : 50 Question Id : 6406531040762 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider the following Java program.

```
class A {
    protected int x, y;
    public A() { x = y = 0; }
    public void setX(int val) { x = val; }
    public void setY(int val) { y = val; }
    public void print() {
        System.out.println("x = " + x + ", y = " + y);
    }
}

class B extends A {
    protected int y;
    public void setY(int val) { y = val; }
}

class C extends B {
    public void setX(int val) { x = (val + 5); }
    public void setY(int val) {
        y = val * 2;
    }
}

class Main {
    public static void main(String[] args) {
        C obj = new C();
        obj.setX(10);
        obj.setY(20);
        obj.print();
    }
}
```

The above program generates output as `x = 15, y = 0`, instead `x = 15, y = 40`, since the inherited `A::y` is overridden in the descendant `B` that hides the instance variable `y` in `A`. Identify the type of anomaly/fault in the given scenario.

Options :

6406533515871. ✖ Inconsistent type use

6406533515872. ✖ State definition anomaly

6406533515873. ✔ State definition inconsistency anomaly

6406533515874. ✖ State visibility anomaly

Question Number : 51 Question Id : 6406531040763 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider the following code segment in Java.

```
class Calculator {
    public double calculate(double num1, double num2) {
        return num1 + num2;
    }
    public int calculate(int num1, int num2) {
        return num1 * num2;
    }
}
```

The class Calculator is mutated as follows:

```
class Calculator {
    public double calculate(double num1, double num2) {
        return num1 * num2;
    }
    public int calculate(int num1, int num2) {
        return num1 + num2;
    }
}
```

Which kind of mutation operator is applied in this case?

Options :

6406533515875. ✔ Overloading Method Change (OMC)

6406533515876. ✖ Argument Number Change (ANC)

6406533515877. ✖ Argument Order Change (AOC)

6406533515878. ✖ Declared/Parameter Type Change (DTC/PTC)

Question Number : 52 Question Id : 6406531040764 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Given a context free grammar over a finite alphabet $\sum = \{a, b\}$, with the production rules as follows:

$$S \rightarrow aSa,$$
$$S \rightarrow BB,$$
$$B \rightarrow b.$$

Let S be the starting variable. Which of the following sets below corresponds to the language generated by the given grammar?

Options :

6406533515879. ✖ $\{a^n bba^n \mid n \geq 1\}$

6406533515880. ✔ $\{a^n bba^n \mid n \geq 0\}$

6406533515881. ✖ $\{a^n (b)^{2n} a^n \mid n \geq 0\}$

6406533515882. ✖ $\{a(b)^{2n} a \mid n \geq 0\}$

Question Number : 53 Question Id : 6406531040770 Question Type : MCQ
Correct Marks : 5

Question Label : Multiple Choice Question

Consider the following control flow graph (CFG).

- $G = \{V, E\}$,
- Set of nodes, $V = \{1, 2, 3, 4, 5, 6\}$,
- Initial node, $V_0 = \{1\}$,
- Final node, $V_f = \{6\}$,
- Set of edges, $E = \{(1, 2), (2, 3), (2, 6), (3, 4), (3, 5), (4, 5), (5, 2)\}$.

Consider the following with respect to the CFG given above:

- Variable x is defined at nodes 1 and 4.
- Variable x is used at nodes 3, 5, and 6.

Which of the following is **not** a valid du-path for the variable x?

Options :

6406533515899. ✖ [1, 2, 3]

6406533515900. ✖ [1, 2, 3, 5]

6406533515901. ✔ [1, 2, 3, 4, 5]

6406533515902. ✖ [4, 5, 2, 6]